

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

df =pd.read_csv("/content/Dataset (1).csv")

df.head(20)
```



	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(P)
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswana Pula(P)
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswana Pula(P)
5	18189371	Din Tai Fung	162	Mandaluyong City	Ground Floor, Mega Fashion Hall, SM Megamall, ...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056314	14.583764	Chinese	...	Botswana Pula(P)
6	6300781	Buffet 101	162	Pasay City	Building K, SM By The Bay, Sunset Boulevard, M...	SM by the Bay, Mall of Asia Complex, Pasay City	SM by the Bay, Mall of Asia Complex, Pasay Cit...	120.979667	14.531333	Asian, European	...	Botswana Pula(P)
7	6301290	Vikings	162	Pasay City	Building B, By The Bay, Seaside Boulevard, Mal...	SM by the Bay, Mall of Asia Complex, Pasay City	SM by the Bay, Mall of Asia Complex, Pasay Cit...	120.979333	14.540000	Seafood, Filipino, Asian, European	...	Botswana Pula(P)
8	6300010	Spiral - Sofitel Philippine Plaza Manila	162	Pasay City	Plaza Level, Sofitel Philippine Plaza Manila, ...	Sofitel Philippine Plaza Manila, Pasay City	Sofitel Philippine Plaza Manila, Pasay City, P...	120.980090	14.552990	European, Asian, Indian	...	Botswana Pula(P)
9	6314987	Locavore	162	Pasig City	Brixton Technology Center, 10 Brixton Street, ...	Kapitolyo	Kapitolyo, Pasig City	121.056532	14.572041	Filipino	...	Botswana Pula(P)
10	6309903	Silantro Fil-Mex	162	Pasig City	75 East Capitol Drive, Kapitolyo, Pasig City	Kapitolyo	Kapitolyo, Pasig City	121.057916	14.567689	Filipino, Mexican	...	Botswana Pula(P)
11	6309455	Mad Mark's Creamery & Good Eats	162	Pasig City	23 East Capitol Drive, Kapitolyo, Pasig City	Kapitolyo	Kapitolyo, Pasig City	121.060820	14.570849	American, Ice Cream, Desserts	...	Botswana Pula(P)
12	6318433	Silantro Fil-Mex	162	Quezon City	Second Floor, UP Town Center,	UP Town Center, Diliman,	UP Town Center, Diliman, Quezon City	121.075419	14.649503	Filipino, Mexican	...	Botswana Pula(P)

					Katipunan Avenue...	Quezon City	Quezon City, Quezon City																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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20 rows × 21 columns



df.info()



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Restaurant ID          9551 non-null   int64
1   Restaurant Name        9551 non-null   object
2   Country Code           9551 non-null   int64
3   City                   9551 non-null   object
4   Address                 9551 non-null   object
5   Locality                9551 non-null   object
6   Locality Verbose       9551 non-null   object
7   Longitude              9551 non-null   float64
8   Latitude               9551 non-null   float64
9   Cuisines                9542 non-null   object
10  Average Cost for two    9551 non-null   int64
11  Currency                9551 non-null   object
12  Has Table booking       9551 non-null   object
13  Has Online delivery     9551 non-null   object
14  Is delivering now       9551 non-null   object
15  Switch to order menu   9551 non-null   object
16  Price range            9551 non-null   int64
17  Aggregate rating       9551 non-null   float64
18  Rating color           9551 non-null   object
19  Rating text            9551 non-null   object
20  Votes                  9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

df.describe()

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggregate rating	Votes
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	2.666370	156.909748
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	1.516378	430.169145
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.000000	0.000000
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	2.500000	5.000000
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	3.200000	31.000000
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.700000	131.000000
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	4.900000	10934.000000

```
df.duplicated().sum()
```

```
0
```

```
df.isna().sum()
```

```
Restaurant ID      0
Restaurant Name    0
Country Code       0
City               0
Address            0
Locality           0
Locality Verbose   0
Longitude          0
Latitude           0
Cuisines           9
Average Cost for two 0
Currency           0
Has Table booking  0
Has Online delivery 0
Is delivering now  0
Switch to order menu 0
Price range        0
Aggregate rating    0
Rating color       0
Rating text        0
Votes              0
dtype: int64
```

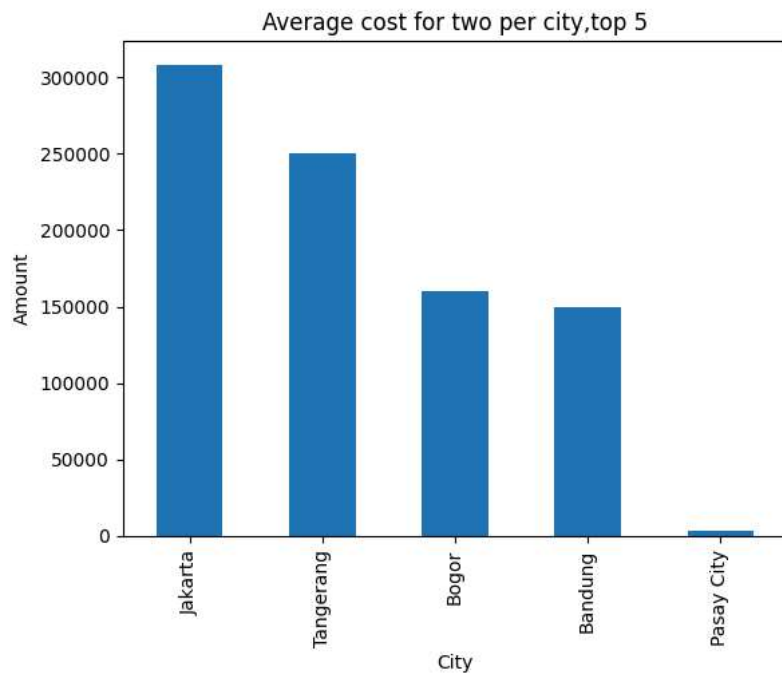
```
df.columns
```

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Is delivering now', 'Switch to order menu',
       'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
       'Votes'],
      dtype='object')
```

```
# group by city ans calculate average cost
```

```
df.groupby("City")["Average Cost for two"].mean().sort_values(ascending=False).head().plot(kind="bar")
plt.title("Average cost for two per city,top 5")
plt.xlabel("City")
plt.ylabel("Amount")

plt.show()
```



```
#cuisines total number of votess  
data=df.groupby("Cuisines")["Votes"].sum().reset_index()
```

```
data[data["Votes"] ==0]
```



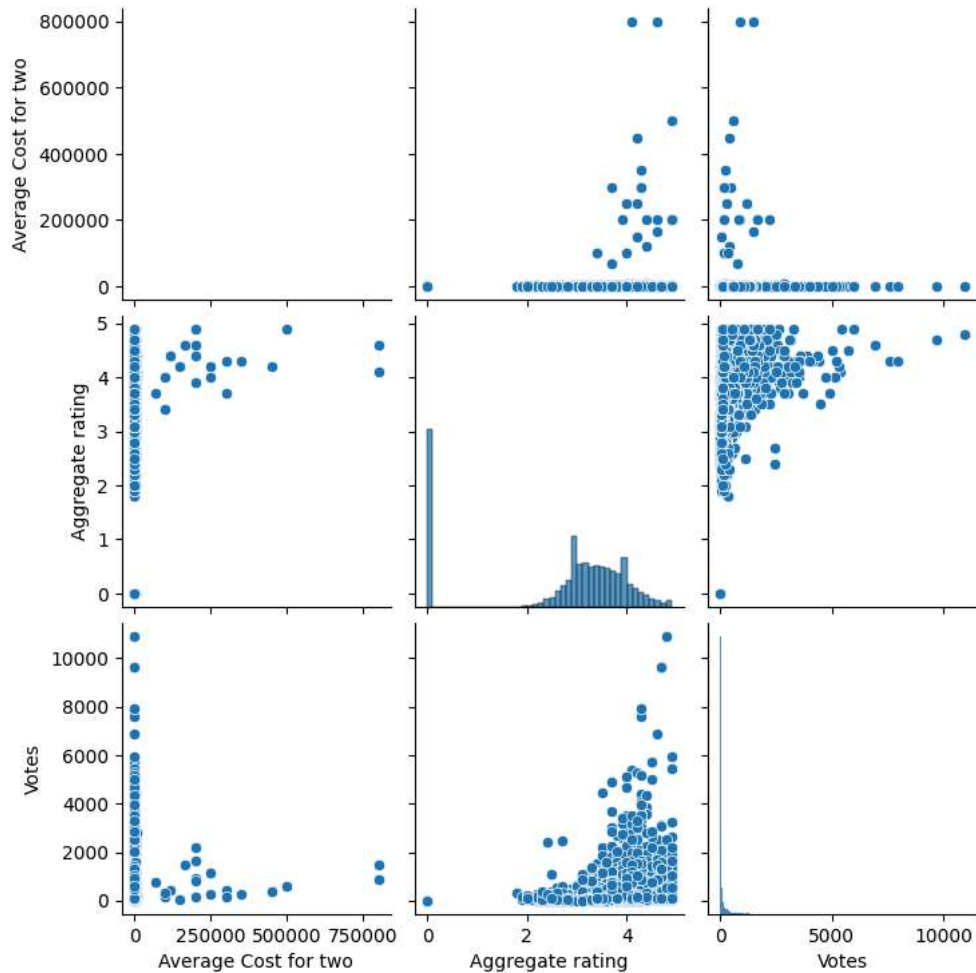
	Cuisines	Votes
2	Afghani, North Indian	0
164	Awadhi, Mughlai	0
185	Bakery, Chinese, North Indian	0
193	Bakery, Desserts, Fast Food, Mithai	0
208	Bakery, Healthy Food	0
213	Bakery, North Indian	0
214	Bakery, Pizza	0
224	Bengali, Chinese	0
234	Beverages, Healthy Food	0
323	Burger, Pizza, Fast Food	0
457	Cafe, Mexican, Continental	0
463	Cafe, North Eastern	0
489	Cafe, Tibetan	0
524	Chinese, Fast Food, Pizza	0
527	Chinese, Italian	0
548	Chinese, North Eastern	0
705	Continental, Lebanese, Mexican	0
844	Fast Food, Chinese, Burger	0
850	Fast Food, Desserts, Beverages	0
863	Fast Food, Mughlai	0
864	Fast Food, Mughlai, Armenian	0
885	Fast Food, South Indian, Mithai	0
983	Ice Cream, Bakery	0
1045	Italian, Continental, American	0
1067	Italian, Fast Food, South Indian	0
1180	Lebanese, North Indian, Fast Food	0
1184	Lucknowi, Mughlai, North Indian	0
1252	Mithai, North Indian, South Indian, Chinese, S...	0
1259	Mithai, South Indian, Chinese, Street Food	0
1299	Nepalese, Tibetan	0
1365	North Indian, Chinese, Italian, Thai	0
1398	North Indian, Chinese, Tibetan	0
1468	North Indian, Indian	0
1509	North Indian, Mithai, Chinese	0
1515	North Indian, Mughlai, Armenian	0
1550	North Indian, Seafood	0
1561	North Indian, South Indian, Chinese, Fast Food...	0
1563	North Indian, South Indian, Chinese, Healthy F...	0
1580	North Indian, South Indian, Italian, Chinese	0
1582	North Indian, South Indian, Mithai	0
1630	Pizza, Ice Cream	0
1656	Raw Meats, Street Food	0
1687	Seafood, Mughlai, North Indian	0
1710	South Indian, Chinese, Mithai	0
1711	South Indian, Chinese, Mughlai	0
1732	South Indian, Pizza, Chinese	0
1811	Tibetan, South Indian, North Indian	0
1816	Turkish, Arabian, Moroccan, Lebanese	0

```
df.columns
```

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
      'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
      'Average Cost for two', 'Currency', 'Has Table booking',
      'Has Online delivery', 'Is delivering now', 'Switch to order menu',
      'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
      'Votes'],
      dtype='object')
```

```
sns.pairplot(df[["Average Cost for two","Aggregate rating","Votes"]])
```

```
<seaborn.axisgrid.PairGrid at 0x7926fb40dde0>
```



```
df.columns
```

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
      'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
      'Average Cost for two', 'Currency', 'Has Table booking',
      'Has Online delivery', 'Is delivering now', 'Switch to order menu',
      'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
      'Votes'],
      dtype='object')
```

```
df["Has Online delivery"].value_counts().plot(kind="pie")
plt.title("Has OnlineDelivery?")
plt.ylabel("")
plt.legend()
plt.show()
```



Has OnlineDelivery?



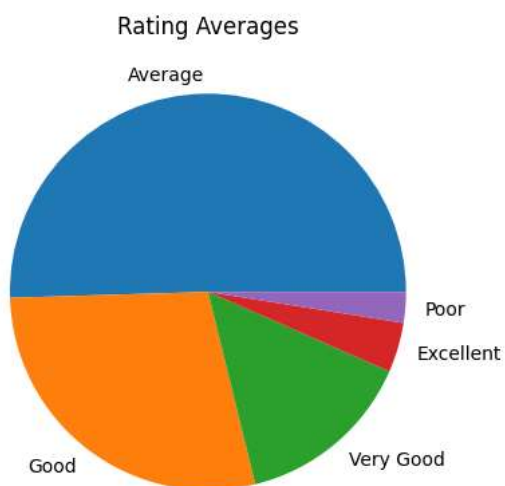
```
df[["Aggregate rating","Rating text"]].head().sort_values(by="Aggregate rating",ascending=True).head(7000)
```



	Aggregate rating	Rating text
2	4.4	Very Good
1	4.5	Excellent
0	4.8	Excellent
4	4.8	Excellent
3	4.9	Excellent

```
df=df[df["Rating text"]!="Not rated"]
```

```
df["Rating text"].value_counts().plot(kind="pie")
plt.title("Rating Averages")
plt.ylabel("")
plt.show()
```



Double-click (or enter) to edit

```
df.columns
```



```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Is delivering now', 'Switch to order menu',
       'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
       'Votes'],
      dtype='object')
```

```
X=df[['Average Cost for two' 'Has Table booking' 'Has Online delivery' 'Price range']]
#Above 2 below 2.5 poor
#Above 2.5 below 3.5 Average
#Above 3.5 below 4.0 Good
#Above 4 below 4.5 Very Good
#Above 4.5 Excellent
```

```
Y=df[["Aggregate rating"]]
```

X



	Average Cost for two	Has Table booking	Has Online delivery	Price range
0	1100	Yes	No	3
1	1200	Yes	No	3
2	4000	Yes	No	4
3	1500	No	No	4
4	1500	Yes	No	4
...	...	...	...	...
9546	80	No	No	3
9547	105	No	No	3
9548	170	No	No	4
9549	120	No	No	4
9550	55	No	No	2

7403 rows × 4 columns

```
from sklearn import preprocessing
LabelEncoder =preprocessing.LabelEncoder()
```

```
X["Has Table booking"] = LabelEncoder.fit_transform(X["Has Table booking"])
```



```
<ipython-input-43-0dc0b67260be>:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus

```
X["Has Table booking"] = LabelEncoder.fit_transform(X["Has Table booking"])
```

X



	Average Cost for two	Has Table booking	Has Online delivery	Price range
--	----------------------	-------------------	---------------------	-------------